Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

1.2 Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. 2. Процесс выполнения лабораторной работы

2.1 Перенаправление ввода-вывода

```
mudzhuraboeva@mudzhuraboeva:~$ ls /etc/ > file.txt
mudzhuraboeva@mudzhuraboeva:~$ ls >> file.txt
mudzhuraboeva@mudzhuraboeva:~$ ls >> file.txt
mudzhuraboeva@mudzhuraboeva:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
```

Рисунок 1: Запись в файл

2.2 Работа фильтра

```
mudzhuraboeva@mudzhuraboeva:~$
mudzhuraboeva@mudzhuraboeva:~$ grep .conf file.txt > conf.txt
mudzhuraboeva@mudzhuraboeva:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
```

Рисунок 2: Поиск расширения .conf

```
/home/mudzhuraboeva/.npm/_cacache/index-v5/1a/ce
/home/mudzhuraboeva/.npm/_cacache/index-v5/c4
/home/mudzhuraboeva/.npm/_cacache/index-v5/1d/f1/ce5e4168fa5f662c23cbbd558c5d82ef5e57d137e2d329fbc17126408620
/home/mudzhuraboeva/.npm/_cacache/index-v5/b7/b7/c3ca4adcf90689fe86f644ec7d227ef4883a4f4b27b859846ec38dcb9b6c
/home/mudzhuraboeva/.npm/_cacache/index-v5/68/ca
/home/mudzhuraboeva/.npm/_cacache/index-v5/9e/e0/c3386deeef310820ca85762b73df78a4cca889774c30364b036d1f870889
/home/mudzhuraboeva/.npm/_cacache/index-v5/c7
/home/mudzhuraboeva/.npm/_cacache/index-v5/c7/b3/c81fcd46b7c6e984952f72b5d5823025eb825b31eb9b584107047503dd5a
/home/mudzhuraboeva/.npm/_cacache/index-v5/f3/6c/c3986891a2eed646eaa7d914d91cffc8d6421272c959bdb5a87486645945
/home/mudzhuraboeva/.npm/_cacache/index-v5/84/ce
/home/mudzhuraboeva/.npm/_cacache/index-v5/ce
/home/mudzhuraboeva/.npm/_cacache/index-v5/09/c0
/home/mudzhuraboeva/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ced5cf6fc29def30220242c53
/home/mudzhuraboeva/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58996cfad55fb2000f59e2edcc
/home/mudzhuraboeva/.npm/_cacache/index-v5/cd
/home/mudzhuraboeva/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d1350482a1d990504757
/home/mudzhuraboeva/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b53cea310cda0b203217f693
/home/mudzhuraboeva/.npm/_cacache/index-v5/c9
/home/mudzhuraboeva/.npm/_cacache/index-v5/af/cf
/home/mudzhuraboeva/.npm/_cacache/index-v5/c0
/home/mudzhuraboeva/.npm/_cacache/index-v5/c3
/home/mudzhuraboeva/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f2506214f94db8861160d3
/home/mudzhuraboeva/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c8188a44d554f3f3ba2e
/home/mudzhuraboeva/.npm/_cacache/index-v5/ca
/home/mudzhuraboeva/.npm/_cacache/index-v5/cf
/home/mudzhuraboeva/.npm/_cacache/index-v5/c1
/home/mudzhuraboeva/snap/hugo/common
/home/mudzhuraboeva/snap/hugo/current
/home/mudzhuraboeva/conf.txt
mudzhuraboeva@mudzhuraboeva:~$
```

Рисунок 3: Поиск файлов

2.4 Команда поиска

```
/etc/foomatic/hashes.d/hashes.new
find: ‘/etc/libvirt’: Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: ‘/etc/lvm/archive’: Permission denied
/etc/logrotate.d/httpd
find: ‘/etc/lvm/backup’: Permission denied
find: ‘/etc/lvm/cache’: Permission denied
find: ‘/etc/lvm/devices’: Permission denied
find: ‘/etc/nftables’: Permission denied
find: ‘/etc/openvpn/client’: Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: ‘/etc/openvpn/server’: Permission denied
find: ‘/etc/pki/rsyslog’: Permission denied
find: ‘/etc/polkit-1/localauthority’: Permission denied
find: ‘/etc/polkit-1/rules.d’: Permission denied
find: ‘/etc/sos/cleaner’: Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: ‘/etc/ssh/sshd_config.d’: Permission denied
find: ‘/etc/sssd’: Permission denied
find: ‘/etc/sudoers.d’: Permission denied
find: ‘/etc/userdb’: Permission denied
/etc/sysconfig/htcacheclean
:
```

Рисунок 4: Поиск файлов

```
mudzhuraboeva@mudzhuraboeva:~$
mudzhuraboeva@mudzhuraboeva:~$ find ~ -name "log*" > logfile &
[1] 6834
mudzhuraboeva@mudzhuraboeva:~$ rm logfile
[1]+  Done                    find ~ -name "log*" > logfile
mudzhuraboeva@mudzhuraboeva:~$
```

Рисунок 5: Фоновый запуск процесса

```
mudzhuraboeva@mudzhuraboeva:~$
mudzhuraboeva@mudzhuraboeva:~$ gedit &
[1] 6866
mudzhuraboeva@mudzhuraboeva:~$ ps | grep gedit
   6866 pts/2    00:00:00 gedit
mudzhuraboeva@mudzhuraboeva:~$ kill 6866
[1]+  Terminated              gedit
mudzhuraboeva@mudzhuraboeva:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

2.7 Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual  page  documents the GNU version of df.  df displays the amount of space available on the
       file system containing each file name argument.  If no file name is given, the space available on  all
       currently  mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environ-
       ment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

       If an argument is the absolute file name of a device node containing a mounted file system,  df  shows
       the  space  available  on  that file system rather than on the file system containing the device node.
       This version of df cannot show the space available on unmounted file systems, because on most kinds of
       systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
       Show information about the file system on which each FILE resides, or all file systems by default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
              scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes;
              see SIZE format below

       --direct
              show statistics for a file instead of mount point

 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

2.8 Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent  sizes rather than device usage; although the apparent size is usually smaller,
              it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and
              the like

       -B, --block-size=SIZE
              scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes;
              see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'

       -c, --total
              produce a grand total
 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

2.9 Команды df и du

```
mudzhuraboeva@mudzhuraboeva:~$
mudzhuraboeva@mudzhuraboeva:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 88846832  65720720  58% /
devtmpfs         4011284        0   4011284   0% /dev
tmpfs            4035660      100   4035560   1% /dev/shm
tmpfs            1614264     1968   1612296   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs            4035660     1640   4034020   1% /tmp
/dev/nvme0n1p3 155186176 88846832  65720720  58% /home
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop1        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop0         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             807132      184    806948   1% /run/user/1008
/dev/sr0         2677920  2677920         0 100% /run/media/mudzhuraboeva/Fedora-WS-Live-43
mudzhuraboeva@mudzhuraboeva:~$
```

Рисунок 9: Справка по команде du

2.10 Команды df и du

```
4        ./.npm/_cacache/index-v5/c1
4        ./.npm/_cacache/index-v5/94/74
4        ./.npm/_cacache/index-v5/94
4        ./.npm/_cacache/index-v5/c1/df
4        ./.npm/_cacache/index-v5/c1
4        ./.npm/_cacache/index-v5/30/1a
4        ./.npm/_cacache/index-v5/30
424      ./.npm/_cacache/index-v5
114660   ./.npm/_cacache
114724   ./.npm
12       ./snap/hugo/25676/.config/go/telemetry/local
0        ./snap/hugo/25676/.config/go/telemetry/upload
12       ./snap/hugo/25676/.config/go/telemetry
12       ./snap/hugo/25676/.config/go
12       ./snap/hugo/25676/.config
12       ./snap/hugo/25676
0        ./snap/hugo/common
16       ./snap/hugo
16       ./snap
0        ./monthly
0        ./reports/monthly/monthly
0        ./reports/monthly
0        ./reports
4        ./ski.plases/equipment
0        ./ski.plases/plans
4        ./ski.plases
0        ./australia
0        ./play/games/play
0        ./play/games
0        ./play
1439928  .
mudzhuraboeva@mudzhuraboeva:~$
```

Рисунок 10: Запуск команды du

2.11 Команда поиска

```
/home/mudzhuraboeva/.npm/_cacache/index-v5/b6
/home/mudzhuraboeva/.npm/_cacache/index-v5/b6/b7
/home/mudzhuraboeva/.npm/_cacache/index-v5/2b
/home/mudzhuraboeva/.npm/_cacache/index-v5/2b/8e
/home/mudzhuraboeva/.npm/_cacache/index-v5/cf
/home/mudzhuraboeva/.npm/_cacache/index-v5/cf/b1
/home/mudzhuraboeva/.npm/_cacache/index-v5/94
/home/mudzhuraboeva/.npm/_cacache/index-v5/94/74
/home/mudzhuraboeva/.npm/_cacache/index-v5/c1
/home/mudzhuraboeva/.npm/_cacache/index-v5/c1/df
/home/mudzhuraboeva/.npm/_cacache/index-v5/30
/home/mudzhuraboeva/.npm/_cacache/index-v5/30/1a
/home/mudzhuraboeva/snap
/home/mudzhuraboeva/snap/hugo
/home/mudzhuraboeva/snap/hugo/25676
/home/mudzhuraboeva/snap/hugo/25676/.config
/home/mudzhuraboeva/snap/hugo/25676/.config/go
/home/mudzhuraboeva/snap/hugo/25676/.config/go/telemetry
/home/mudzhuraboeva/snap/hugo/25676/.config/go/telemetry/local
/home/mudzhuraboeva/snap/hugo/25676/.config/go/telemetry/upload
/home/mudzhuraboeva/snap/hugo/common
/home/mudzhuraboeva/monthly
/home/mudzhuraboeva/reports
/home/mudzhuraboeva/reports/monthly
/home/mudzhuraboeva/reports/monthly/monthly
/home/mudzhuraboeva/ski.plases
/home/mudzhuraboeva/ski.plases/equipment
/home/mudzhuraboeva/ski.plases/plans
/home/mudzhuraboeva/australia
/home/mudzhuraboeva/play
/home/mudzhuraboeva/play/games
/home/mudzhuraboeva/play/games/play
mudzhuraboeva@mudzhuraboeva:~$
```

Рисунок 11: Поиск директорий

3. 3. Выводы по проделанной работе

3.1 Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.